

Image credit: Jenny List on Hackaday

Abstract

This document explains piezoelectric quartz crystal oscillators. It covers various information about the applications, how they work, and their history. When relevant, it also includes some information about the broader subject containing quartz oscillators, which are piezoelectric components.

Applications

Crystal oscillators have a very broad range of applications. A partial list includes computers, modems, LiDAR, radio, communications systems, microcontrollers, watches, and clocks. However, the most general use is for electrical timing. Quartz oscillators give out a very consistent alternating voltage waveform, making them ideal for precise timing. Electrical devices harness this wave and count time in clock cycles, “ ticks,” to manage their operations. People often use a beating heart as an analogy to describe its function. There are several applications for quartz oscillators, including data management, frequency matching, and signal stabilization. all three rely on their precise time keeping ablility.



Fig. 1. Circuit symbol for a crystal oscillator



Fig. 2. first quartz watch [1]

The simplest example is a quartz watch. Watch manufacturers take advantage of this component to create super low-maintenance pieces, hence the name quartz watch. Quartz watch pieces are meant to be extra precise. The first quartz watches hit the market in 1969 [1].

A more important example is for computers. Because computers run on binary logic, the sinusoidal wave produced by the crystal needs to be converted into a square wave before being used. Inside a computer, memory in its simplest form is stored in circuits called flip-flops and latches. These circuits are wired directly to the clock. The clock ticks make sure the data flows in unison; if not, important numbers will become corrupted, and calculations will return incorrect answers. In addition, most components need a given amount of time to change states. For example, if we attempt to access data while a transistor is switching, we will not receive proper information. A good analogy for quartz oscillators is a metronome keeping the beat for musicians; it would sound bad off beat.

Piezoelectric effect

A special property of some crystals is piezoelectricity; one of the strongest exhibitors of the effect is quartz. As explained by Lord Kelvin[2], “simple elongation in any direction perpendicular to the principal axis of a quartz crystal produces electro-polarization.” Polarization is when one side of an object is negatively charged, and the other is positively charged. The axis depends on the molecular structure of the quartz. This also works with compression; the polarization will just be opposite in sign to what it would be if elongated.

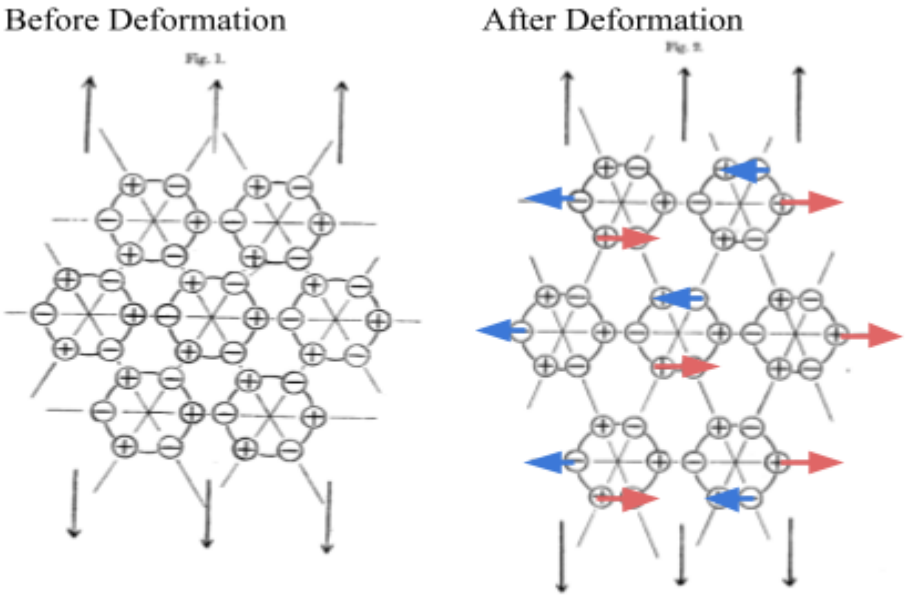


Fig. 3. Annotated diagram by Lord Kelvin[2]. Blue shows negative charge paths and red shows positive. All charges are moving, and in the same direction, even though only a few of their paths are drawn. Kelvin drew the difference in a way difficult to tell.

As observed by Lord Kelvin (Fig. 3), the reason why quartz would polarize under pressure was due to the chemistry of quartz and how the atoms in the crystal move. Quartz is formed when oxygen and silicon atoms combine into a repeating hexagonal shape called a lattice. When this happens, the oxygen atoms take electrons from the silicon atoms, becoming negative. When the silicon donates electrons to the oxygen, it becomes positive. When pressure is applied, the lattice deforms. The negative oxygen and positive silicon are pushed in opposite directions, so one side becomes negative and the other positive.

When we attach electrodes to the quartz, this converts the polarization into a voltage. An example of this being used is in lighters: when you press the igniter, it presses a quartz crystal and uses the voltage to make a fuel-igniting spark. To quickly summarize, deformation causes polarization, which then makes voltage. Similar to other electrical phenomena, this process can be done in the exact reverse. When we apply a voltage to the crystal electrodes instead of receiving, this causes the reverse piezoelectric effect. It deforms the quartz without us pressing it. When the reverse effect is used, it causes electro-mechanical oscillations, resulting in the voltage sine waves our devices use for timing.

In order to model the crystals' electro-mechanical nature in a fully electrical nature, we need to use an RCL circuit as described in a journal from Microchip [3]. We say the components of the RCL describe the “motional” resistance, inductance, and capacitance of the crystal. where all three components are in series. There is also a second capacitor in parallel to describe the electrodes on the crystal, which act as a capacitor. The key difference between a real RCL and a crystal circuit, however, is that the crystal generates a much more reliable waveform.



Fig. 4. ABLS7M2 [4]



Fig. 5. Quarter and ABLS7M2 size comparison, (generated with comparesizes.com)

Example Component

As an example of quartz oscillators, we will examine the datasheet for the Abracon ABLS7M2 [4]. This component doesn’t have any unusual specifications, so it will give a good idea about what to expect from a random oscillator. The part can oscillate at frequencies between 12 and 40 megahertz. Inside that range, it also has a list of standard frequencies it's designed to operate at, ranging from 12 to 27 megahertz. The listed frequencies are what engineers behind the component guarantee work best. There are several types of cuts you can give to a crystal, which give it different properties. This is

AT-cut, which is the most common kind today. Cuts are specified by their angles in relation to the axis of the crystal lattice. There is also information about how ageing and temperature can affect the crystal, the effects of which can change the frequency by a few hertz. The component has some internal resistance and capacitance across it, which are listed in the data sheet. The size is tiny, only 7 x 4.1 x 2 mm. For comparison, a quarter's surface is sixteen times larger.

History

The root piezo is from piezein, a Greek word meaning to press[5]. The company American Piezo is a manufacturer of various piezoelectric elements and posted a blog retelling the history of the effect[5] from its discovery until the 1960s, retold in this segment.

Piezoelectricity was discovered by the Curie brothers, Pierre and Jacques, who, after researching a similar pyroelectric effect (voltage by temperature change in a crystal), wondered if it was possible to create voltage through pressure. They searched for the effect in materials including sugar, topaz glue, and tinfoil, and eventually discovered the effect inside Rochelle salt and quartz. The brothers went on to build the first piezoelectric electrometer, a device that specializes in measuring very small voltages. The brothers also discovered the reverse piezoelectric effect later on. Lord Kelvin[2] was also an early investigator of the effect, whose findings were discussed earlier. His ideas were foundational for piezoelectric theory.

During and before WWI, piezoelectricity was already being investigated by inventors. One French scientist, Paul Langévien, was able to use it to create the first submarine sonar with a device called a quartz sandwich transducer. Another man, Alexander Nicolson, independently created a similar system. Nicolson was known for inventing early sonars and ultrasonic devices with piezoelectrics[1]; he also used them to make speakers and microphones. In 1917, Nicolson invented the first crystal oscillator, but instead of quartz it used Rochelle salt.

Later, Walter G. Cady invented the first actual quartz oscillator[6]. He filed his invention for patent on May 28th, 1921. It became U.S. patent 1,472,583, titled Method of Maintaining Electric Currents of Constant Frequency. Cady was a German immigrant who got his degree from the University of Berlin. Upon arriving in the US, he worked on anti-submarine devices like Langévien and Nicolson did [1]. Later on, he had the idea to use piezoelectricity for radio frequency components. He initially wanted to use Rochelle salt; however later moved to quartz. The reason for the change was that,

> “When applying an AC voltage over a wide range of frequencies, Cady noticed that both Rochelle salt and quartz crystals would reach their maximum amplitude oscillation (resonance) over a very narrow range of frequencies. Quartz,
> however, had a sharper resonance curve than Rochelle salt.”[1]

Previously, radio frequency standards were based on the resonance curve of a tuning fork, but Cady noticed that the curves of quartz and tuning forks were nearly identical. Cady filed his patent, believing it had the potential to replace tuning forks. Crystal oscillators were crucial in early continuous wave-type radios. Old radios also used alternator or vacuum tube oscillators to generate continuous waves. Quartz oscillators replaced tuning forks, vacuum tubes, and alternators for radio purposes. “The most critical component of a radio receiver or transmitter is that which controls the operating frequency.”[7].

Due to the similarities between Nicolson and Cady’s inventions, a legal battle ensued. “Subsequent litigation [between Cady and Nicolson for patent infringement] resulted in a legal decision in favor of Nicolson”[8]. Because of this, Nicolson is usually cited as the inventor of Piezoelectric oscillators in general, but Cady is cited as the inventor of quartz oscillators specifically. This often causes confusion. The thing to note is that all quartz oscillators are crystal oscillators; however, crystal oscillators are not all quartz oscillators. It can also be observed that Cady is mentioned more often than Nicolson in historical articles, which is due to the importance of quartz oscillators over general crystal ones.

The historical details from WWII and after have been detailed by Professor R. J. Thompson[7]. At this time, quartz radios were becoming popular among the public. The war and improvements in radio due to quartz pushed the US military to adopt wireless communications. The demand from both led to a significant shortage of oscillators; however, the pressure led manufacturers to improve the oscillators even more, creating new methods of production to increase their outputs, soon fixing the shortage. Soon after the shortage ended, a second crisis emerged, where crystals seemed to age and fall out of tune or stop oscillating, called the ageing crisis. It was found that this was caused by defects made when grinding the quartz. The resolution of both these events made the resulting crystals of very high quality. Due to the war effort, all the producers of oscillators shared technical details, so these improvements were able to happen very rapidly.

Computers were always using quarts. ENIAC, standing for Electronic Numerical Integrator And Computer, was the first programmable general-purpose computer ever made. It was built during WWII at the University of Pennsylvania to calculate artillery trajectories. Inside, “Timing was controlled by a 100-kHz quartz oscillator, resulting in a basic pulse width of 10 milliseconds.”[9]. ENIAC was a synchronous and parallel machine[9], synchronous because it used controlled pulses to



Fig. 6. two women working on ENIAC [9]

manage operations, and parallel because it could utilize different arithmetic units simultaneously. Doing this without precise timing would have been almost impossible. It is an excellent example of the importance of oscillators.

Social impacts

Quartz oscillators alone did not have a profound impact on society; however, the devices they made possible did. The first, most important thing was the radio. During the initial radio boom of the 1920s, quartz was (with few exceptions) not used; instead, vacuum tubes were used for oscillation.

Radio provided the nation with common access to entertainment, news, and music. During the boom, radio had “blunted regional differences and imposed similar tastes and lifestyles.”[10]. This strongly shaped American culture and national identity. Radio also had a profound effect on politics, which still exists today. In the modern era, a politician’s personality is a highly important part of a voter's opinion and decision-making, even if subconsciously. The importance of a politician's personality began in the radio boom when, for the first time, people no longer had to read candidates' words but could listen to them in real time.

> “Politicians’ personalities all of a sudden started to matter more. The way their voices sounded made more of a difference. And their ability to engage and entertain became crucial components of their candidacies” [11].



Fig. 7. The first commercial quartz oscillator, the General Radio Type 275. [1]

Although the early radios did not use quartz, this soon changed. The first quartz oscillator people could buy was the General Radio Type 275 [1]. The fact that the first of its kind was dedicated to radio is proof of their importance to eachother. The device hit the market in 1924 at $145; it may seem low, but it was equivalent to about $ 2,350 today. For comparison, an average actual radio set was about $50. The oscillator was also very large, 880 cubic inches, over 250 thousand times larger than modern components.

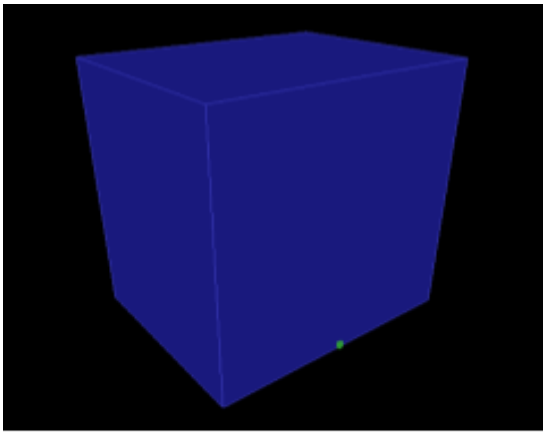


Fig. 8. General Radio Type 257 and ABLS7M2 size comparison, (generated with comparesizes.com)

The sale of the General Radio Type marked the beginning of a radio revolution. Quartz radios gave finer frequency accuracy than others could. Their accuracy became increasingly important as radio regulations were created. In 1927, Congress established the FCC (Federal Communications Commission), which required broadcasters to remain within 500Hz of their intranded frequency [1]. This range is called bandwidth. It later shrunk to 50Hz in 1932 [1]. The key component to allow the shrinking bandwidth was the quartz oscillator, which was already standard by the 1932 regulations. The smaller the bandwidth, the more stations can broadcast, and the more information people have access to. As stations and radio became more abundant due to quartz oscillators and other improvements, the social effects of the radio took more effect. Quartz oscillators and other improvements made radio and radio stations more abundant, strengthening the social effects they had. In a way, this was a prelude to today's information era.

Limits and future possibilities

Today, quartz oscillators are extremely reliable, small, and precise. We have begun using lab-grown quartz instead of natural quartz to ensure quality. The size of an average modern oscillator is in the range of millimeters and can reach frequencies in gigahertz. Their precision is within one tenbillionth of a hertz, which is ten million times better than original radio requirements and two hundred thousand times better than today's requirements [1]. As we get a more precise frequency, the cut and structure of the crystal also need to be more precise. Consequently, the imperfections in crystals make it more and more difficult to reach higher precision. Because no two crystals can be the same, we can't manage the structure any further; we are reaching the limit on precision[1]. Until then, oscillators will continue to gain more precision, smaller footprints, and higher frequencies.

Like how the work on other piezoelectric devices led to the development of oscillators, experts on oscillators and other piezoelectric devices are now developing MEMS, micro-electromechanical systems. The goal is to miniaturize mechanical systems, similar to how we were able to miniaturize circuits into an integrated circuit. People are interested in applying them to the aerospace, automotive, and medical fields[12]. Just like in oscillator components, precision is extremely important here. One example is a shear stress sensor (Fig. 9), which uses capacitive effects to measure the motion of the mechanical components. Unlike other devices used to measure shear stress, this MEMS chip allows people to measure shear stress directly, making it easier and more reliable to use.

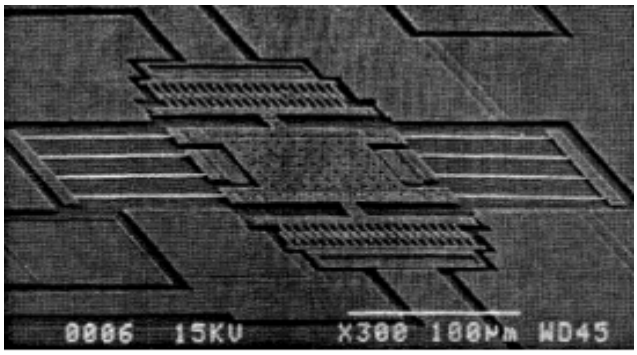


Fig. 9. shear stress sensor MEMS chip [12]

A new, precise method for time keeping is now being investigated for quantum computing. A new form of matter called a time crystal is a structure that is periodic in both space and time, meaning it oscillates naturally without outside help. They may appear to violate the laws of physics; however, no energy is entering or leaving the crystal, so thermodynamics is respected[13]. According to researchers, “Having an object like a time crystal that is stable against experimental interference may help design quantum states that are long-lived, a crucial task for the future improvement of quantum processors” [Xiao Mi 13] and “This is key for future applications of quantum processors and realization of fault-tolerant quantum computation.”[Kostyantyn Kechedzhi 13]. Just like quartz oscillators and regular computers, time crystals are meant to act like a clock for quantum computers. If we can understand how to create them, we can make future quantum computers much more stable. Quantum computers seem very promising; they can perform more calculations than any normal computer ever can by exploiting advanced quantum physics, and thus the importance of time crystals to operate them.

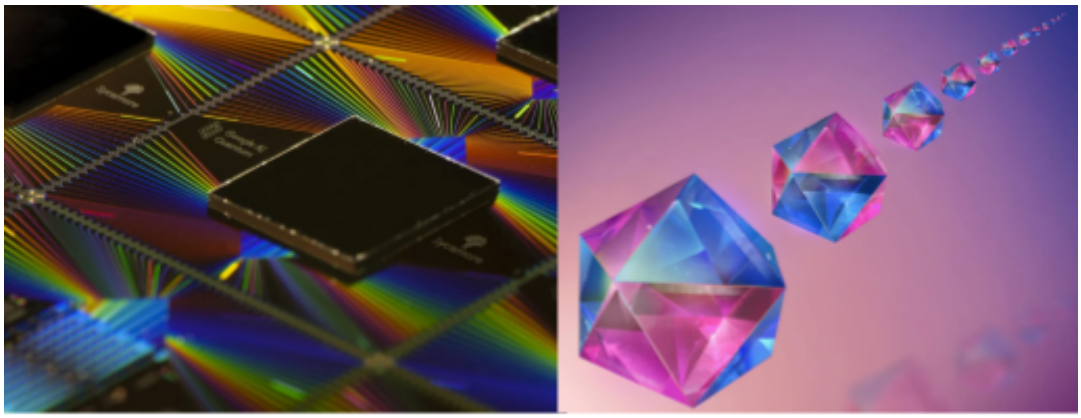


Fig. 10. google's quantum chip sycamore (left) and an artists depiction of time crystals (right) [13]

Conclusion

Piezoelectric quartz oscillators are highly essential to many applications. They may appear niche at first glance; however, they are in several components you interact with daily. Microwaves, computer mice, keypads, and anything wireless all use them. There are many more uses than could have been explained thoroughly. The easiest way to spot them is to look for devices with any microchips, no matter how simple, the odds are it uses one. There are three main applications where you may use oscillators: data movement, signal processing, and frequency matching, all of which rely on the precise timekeeping the oscillator makes possible. When quartz oscillators improve, it improves everything with them. Both the piezoelectrics and the greater electronics industry are strongly related to quartz oscillators.

[1] M. Lombardi, “The evolution of Time Measurement, part 2: Quartz Clocks [recalibration],” *IEEE Instrumentation & Measurement Magazine*, vol. 14, no. 5, pp. 41–48, Oct. 2011. doi:10.1109/mim.2011.6041381

[2] Lord Kelvin, “XXXIII. On the Piezo-Electric Property of Quartz,” *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, vol. 36, no. 221, pp. 331–342, Oct. 1893. doi:10.1080/14786449308620484

[3] S. Bible, AN826 Crystal Oscillator Basics and crystal selection for ... - microchip, https://www.yumpu.com/en/document/view/4711762/an826-crystal-oscillator-basics-and-crystal-selection-for-microchip (accessed Feb. 27, 2026).

[4] Abracon, “SMD MICROPROCESSOR CRYSTAL”, ABLS7M2, [Revised: 29.07.2021-]. [Online-]. Available: https://abracon.com/product-lineup/frequency-control-timing-devices/quartz-crystals/mhz-crystals [Accessed: 02.11.2026-].

[5] “A brief history of piezoelectricity: APC int,” American Piezo, https://www.americanpiezo.com/blog/a-brief-history-of-piezoelectricity/ (accessed Feb. 27, 2026).

[6] W. G. Cady, “Method OF Maintaining Electric Currents OF Constant Frequency,” Oct. 30, 1923

[7] R. J. Thompson, "The development of the quartz crystal oscillator industry of World War II," in IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control, vol. 52, no. 5, pp. 694-697, May 2005, doi: 10.1109/TUFFC.2005.1503957.

[8] V. E. Bottom, A history of the Quartz Crystal Industry in the USA | IEEE UFFC, https://ieee-uffc.org/about-us/history/uffc-s-history/a-history-of-the-quartz-crystal-industry-in-the-usa (accessed Feb. 28, 2026).

[9] H. Neukom, “The Second Life of Eniac,” *IEEE Annals of the History of Computing*, vol. 28, no. 2, pp. 4–16, Apr. 2006. doi:10.1109/mahc.2006.39

[10] Digital history, https://www.digitalhistory.uh.edu/disp_textbook.cfm?smtID=2&psid=3397 (accessed Feb. 27, 2026).

[11] R. Gunderman, “How a 1920s Radio Broadcast Changed Politics Forever - Fast Company,” Fast Company Design, https://www.fastcompany.com/90570860/how-a-1920s-radio-broadcast-changed-politics-forever (accessed Feb. 28, 2026).

[12] M. Mehregany and M. A. Huff, "Microelectromechanical systems," Proceedings IEEE/Cornell Conference on Advanced Concepts in High Speed Semiconductor Devices and Circuits, Ithaca, NY, USA, 1995, pp. 9-18, doi: 10.1109/CORNEL.1995.482415.

[13] C. Q. Choi, K. Kechedzhi, and X. Mi, “ What’s a Time Crystal? And how do Google researchers use quantum computers to make them?,” IEEE Spectrum,